# CAMBRIDGE O LEVEL COMPUTER SCIENCE (2210 / 0478)

Paper 1: Computer Systems | Topic-Wise Question Bank (2024 – 2026 Series)

## TOPIC 01: DATA REPRESENTATION

*Exam Sources: Cambridge 2210 / 0478 (May/June 2023, Oct/Nov 2023, May/June 2024, Oct/Nov 2024)*

### Question 1 [2210/12/M/J/24 Q1 (2024 Series)] [ Total: 6 marks ]

(a) Convert the 8-bit binary number 01011010 to denary. **[1 mark]**

(b) Convert the denary number -45 to an 8-bit binary number using two's complement representation. **[2 marks]**

(c) Perform the binary addition of 01011010 and 00110111. State whether an overflow error has occurred and explain your answer. **[3 marks]**

**✓ Marking Scheme / Model Answer:**

(a) 01011010 = 64 + 16 + 8 + 2 = 90 [1 mark]
(b) +45 in 8-bit binary: 00101101 [1 mark]
One's complement: 11010010, add 1 -> Two's complement: 11010011 [1 mark]
(c) Addition: 01011010 + 00110111 = 10010001 [1 mark]
No overflow error has occurred [1 mark] because the result fits within 8 bits (no carry out from the MSB / result is within 8-bit range 0 to 255) [1 mark].

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### Question 2 [2210/12/M/J/23 Q1 (2023 Series)] [ Total: 6 marks ]

(a) Explain the difference between the ASCII character set and the Unicode character set. **[3 marks]**

(b) A text document contains 2,048 characters encoded in standard 8-bit ASCII. Calculate the file size of the text document in Kibibytes (KiB). Show your working. **[3 marks]**

**✓ Marking Scheme / Model Answer:**

(a) ASCII uses 7 or 8 bits per character, representing up to 128 or 256 unique characters (mainly English letters, digits, punctuation) [1.5 marks].
Unicode uses 16 or 32 bits (e.g. UTF-8 / UTF-16), representing over 1 million characters from all global languages, symbols, and emojis [1.5 marks].
(b) Working: Total bits = 2048 characters * 8 bits = 16,384 bits = 2,048 Bytes [1 mark]
In KiB: 2,048 Bytes / 1,024 = 2 KiB [2 marks]

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## Question 3 [2210/12/O/N/23 Q1 (2023 Series)] [ Total: 5 marks ]

(a) The binary register contains: 0 0 1 1 0 1 0 0 (Denary: 52).

Perform a logical shift of 2 places to the left. Write down the binary contents and state the new denary value. **[2 marks]**

(b) Explain the arithmetic effect of a logical left shift and state what happens if bits are shifted out of the register (overflow/loss of precision). **[3 marks]**

**✓ Marking Scheme / Model Answer:**

(a) Shift 2 places left: 1 1 0 1 0 0 0 0 [1 mark]
Denary value: 128 + 64 + 16 = 208 [1 mark]
(b) Each 1-place left shift multiplies the denary value by 2 (shifting 2 places multiplies by 4: 52 * 4 = 208) [1.5 marks].
If a 1-bit is shifted out of the most significant bit (MSB), an overflow error occurs and the number loses accuracy / precision [1.5 marks].

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## Question 4 [2210/11/M/J/24 Q1 (2024 Series)] [ Total: 5 marks ]

(a) Convert the hexadecimal value 3B to binary and denary. **[2 marks]**

(b) State two reasons why computer scientists use hexadecimal notation instead of binary. **[2 marks]**

(c) State one application of hexadecimal in computer systems. **[1 mark]**

**✓ Marking Scheme / Model Answer:**

(a) Binary: 00111011 [1 mark]. Denary: (3 * 16) + 11 = 59 [1 mark].
(b) Shorter/more compact representation; easier for humans to read/remember; reduces transcription errors [2 marks].
(c) MAC addresses / IPv6 addresses / HTML color codes / Error codes / Memory dump [1 mark].

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